Ensemble Deep Learning Framework for Skin Cancer Detection Using Dermoscopy Images

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***Abstract*— Early melanoma diagnosis and clinical decision support depend heavily on automated skin cancer detection using dermoscopy images and deep learning-based analysis. Current methods that rely on single convolutional neural network models or manual assessment are subjective and have poor generalisation across various lesion patterns. To increase robustness and classification performance, an ensemble deep learning framework combining refined InceptionV3, DenseNet, and EfficientNet with a weighted voting approach is proposed. Experimental testing confirms the proposed ensemble model's efficacy for accurate skin cancer classification, with a validation accuracy of 97.8%, an F1-score of 0.96, and an AUC of 0.98.**

Keywords - Skin Cancer Detection, Dermoscopy Images, Deep Learning, InceptionV3, Ensemble Learning, Weighted Voting.

I. Introduction

One of the most common and quickly spreading types of cancer in the world is skin cancer, which presents a significant public health concern. Melanoma has a significant potential for metastasis, making it the most aggressive and potentially fatal kind of the disease. In clinical practice, prompt and precise diagnosis is essential since early detection of skin cancer greatly increases survival rates. Dermoscopy, a non-invasive imaging method that allows dermatologists to see subsurface features that are invisible to the unaided eye, has become a routine tool for visualising skin lesions.

The World Health Organisation (WHO) reports that millions of new cases of skin cancer are recorded worldwide each year, with melanoma being responsible for over 60,000 fatalities. Research shows that early detection of melanoma increases the survival probability to over 95%, but delayed diagnosis significantly lowers patient survival. Long-term UV exposure, changing lifestyles, and ageing populations are all contributing contributors to the ongoing increase in incidence, underscoring the critical need for accurate screening and diagnosis methods.

Traditional methods of diagnosing skin cancer mostly rely on clinical knowledge, dermoscopic examination, and visual inspection. However, these techniques are intrinsically subjective and rely on dermatologists' expertise, which results in inconsistent diagnoses. Asymmetry, border irregularity, colour fluctuation, and texture are examples of lesion characteristics that might be interpreted inconsistently, leading to misdiagnosis, needless biopsies, or missed malignant cases, all of which raise patient risk and healthcare costs.

Artificial intelligence (AI), especially deep learning, has become a potent tool in medical picture analysis to address these issues. By automatically extracting discriminative features from massive datasets, Convolutional Neural Networks (CNNs) have shown remarkable performance in picture classification tasks.Deep learning-based methods are ideal for automated skin cancer detection using dermoscopic pictures because they provide benefits like scalability, reproducibility, and objectivity.CNN architectures including VGG, ResNet, Inception, DenseNet, EfficientNet, MobileNet, and Vision Transformers are among the deep learning methods that have been investigated for this application. To make use of pretrained models and lessen the requirement for sizable labelled datasets, transfer learning has been extensively used. Furthermore, many models have been combined using ensemble learning approaches, which improve classification accuracy and generalisation by lowering individual model bias.

Existing deep learning techniques have a number of drawbacks despite their success. A lot of models are overfitted and perform worse on untested data since they are trained on little or unbalanced samples. Complex lesion changes may be difficult for single-model techniques to capture, and their high computational complexity and interpretability make them difficult to use in real-world clinical settings. Additionally, problems with data quality including noise, artefacts, and different imaging settings have a detrimental effect on model reliability.In automated skin cancer detection systems, negative characteristics including false positives and false negatives continue to be serious problems. While false negatives may postpone treatment and raise the danger of death, false positives might result in needless anxiety and medical treatments.

The broad use of AI-based diagnostic tools in healthcare settings is further hampered by ethical issues, a lack of standardised datasets, and a lack of clinical validation. This work focusses on an ensemble deep learning architecture for dermoscopy image-based skin cancer detection in order to address these issues.The method attempts to improve robustness, accuracy, and generalisation by combining refined InceptionV3, DenseNet, and EfficientNet models using a weighted voting process.The suggested system is intended to help dermatologists by offering dependable, scalable, and consistent diagnostic support, which will enhance early detection and improve patient outcomes.

Additionally, the growing incorporation of AI-powered diagnostic tools into mobile health platforms and telemedicine underscores the usefulness of automated skin cancer diagnosis. These systems have the potential to expand dermatological treatments to areas with limited resources and access to specialists. Deep learning-based solutions can minimise diagnostic delays, improve healthcare processes, and facilitate large-scale population screening by allowing early screening and prioritisation of high-risk cases. The necessity for precise, resilient, and clinically sound ensemble deep learning models for skin cancer diagnosis is further supported by this expanding applicability.

II. Literature Review

Kadampur et al. [1] evaluated deep learning-based skin cancer classification models using the HAM10000 benchmark dataset and discovered promising diagnosis outcomes. The models' high ROC values showed that they could reliably distinguish between malignant and non-cancerous lesions. SqueezeNet, DenseNet, and InceptionV3 outperformed ResNet in ROC scores among the pretrained models; nevertheless, the study primarily examined comparative performance and did not assess resilience across various clinical situations.To enhance skin cancer identification using dermoscopic images, Akinrinade et al. [2] investigated deep learning techniques by developing an end-to-end classification system. To address the issues of class imbalance and inadequate data, a number of deep learning backbone designs were studied using transfer learning, data augmentation, and Generative Adversarial Networks (GANs).The study also compared ensemble and hybrid models to find the optimal approach for early skin cancer diagnosis.

A CNN-based skin cancer detection model with multiple convolutional layers (32–256 filters), ReLU activation, max-pooling, and Softmax output was created by Senthil Kumar et al. [3] using Keras. With a batch size of 32 and 50 epochs of training on 150x150 images, the model was able to diagnose benign and malignant tumours with an accuracy of 86%. The small dataset size was the reason for the comparatively poorer performance when compared to more sophisticated CNN models.Khamsa et al. [4] filled a gap in earlier research that handled both tasks separately by proposing a unified deep learning architecture that combines skin cancer diagnosis and detection into a single system. Larger datasets, data augmentation, and more recent YOLO versions were found to have the ability to increase the evaluation performance of the approach, which used YOLOv5 for lesion diagnosis. The study showed that the suggested framework can be expanded to identify and diagnose various forms of cancer and used in more general fields including biometric recognition and autonomous systems.

Sheeba Rachel et al. [5] used ResNet-50 and ResNet-101 for feature extraction on a dataset of 660 photos after preprocessing the images using the dull razor approach. Several machine learning techniques were used to classify the retrieved features, and the models were contrasted using VGG and Inception architectures. ResNet-101 with a neural network classifier produced the maximum accuracy of 85.6%, proving the efficacy of residual learning.

Mukesh Kumar et al. [6] highlighted the rising incidence of skin cancer worldwide and the significance of early diagnosis in lowering death and diagnostic expenses. In order to diagnose melanoma, the study used machine learning and deep learning techniques together with ensemble classifiers like AdaBoost and feature extraction techniques like GLCM. The efficacy of the suggested approach in assisting dermatologists and lowering misdiagnosis was confirmed by the experimental findings, which showed a high accuracy of 98.56%. Future improvements are suggested through improved preprocessing and sophisticated modelling techniques.Shereen Moataz Afifi et al. [7] used Flutter to create SkinKo, a cross-platform mobile application for automated skin lesion analysis for iOS and Android smartphones. Through TensorFlow Lite integration, the application enables users to do on-device categorisation, store scan history, and take or upload lesion photos.An efficient mobile-based skin cancer screening solution is demonstrated by the initial classification of lesions as benign, malignant, or healthy, followed by the activation of specialised classification models to identify particular disease kinds.

Kajol Kathuria et al. [8] examined deep learning methods for the identification and categorisation of skin cancer, highlighting the significance of early detection, especially for melanoma. Convolutional Neural Networks (CNNs) are the most efficient models for precise skin lesion classification because of their superior performance in medical image analysis, according to the study, which described a typical workflow encompassing preprocessing, segmentation, feature extraction, and classification.A deep learning-based skin cancer detection system utilising pretrained MobileNetV2 and Xception models for feature extraction and classification was presented by Usha et al. [9]. In order to decrease overfitting and enhance generalisation, the method used data augmentation techniques. A Softmax layer was used to classify lesions into many categories. Experiments on a dataset of 15,069 dermoscopic images showed good performance in terms of accuracy, precision, recall, and F1-score metrics, demonstrating the model's resilience while dealing with unbalanced data. K. Suneetha et al. [10] used k-means segmentation, WEKA-based feature extraction, and hybrid classification to study melanoma classification on the PH2 dataset.The method enhanced melanoma detection accuracy, precision, recall, and sensitivity by combining CNN features with machine learning classifiers and fine-tuning strategies.

Takfarines Guergueb et al. [11] used a sizable dataset created by combining several public sources to compare 20 deep learning models for melanoma diagnosis. According to the study, EfficientNetB7 performed better, with an AUC of 99.01%. Subsequent research brought to light issues with dataset imbalance and the necessity of expanding the methodology to multi-class skin lesion categorisation. Ravi Prakash et al. [12] investigated transfer learning-based skin cancer diagnosis using VGGNet19, ResNet152V2, MobileNetV3Large, and a hybrid transfer learning model on the HAM10000 dataset. The hybrid approach performed better than individual models, with training accuracy of 99.53% and validation accuracy of 97.52%. In clinical skin cancer screening, the study showed how fine-grained feature analysis can reduce false positives and increase diagnostic efficiency.

Mary Linda Panakal Augustine et al. [13] compared models such as InceptionV2, InceptionV3, DenseNet, MobileNet, and VGG19 to explore deep learning-based techniques for accurate skin cancer diagnosis. The study evaluated the benefits and drawbacks of each architecture using evolutionary optimisation techniques to identify the optimal diagnostic model. The findings showed how deep learning systems could improve early detection accuracy, reduce misdiagnosis, and maximise the use of medical resources.Edi Junaedi et al.[14] used the ISIC-2024 dataset to develop an ensemble deep learning system that combines Vision Transformer (ViT), InceptionV3, and MobileNetV2 for the identification of skin cancer. In comparison to individual models and other CNN-based techniques, the ensemble obtained 99.45% accuracy with 100% recall and 98.91% precision, successfully lowering false negatives. The model's accuracy and applicability for clinical skin cancer diagnosis were validated by visual assessments and experimental comparisons.

A CNN-based skin cancer diagnosis model combining VGG16, VGG19, and ResNet50 with transfer learning and the Adam optimiser was proposed by Kshirabdi Tanaya Rath et al. [15]. With an accuracy of 93.50%, the ensemble outperformed previous methods and proved its usefulness for early detection of skin cancer. Future research will concentrate on optimising optimiser settings to further improve performance, as the study demonstrated the model's potential for therapeutic support.Nanthini N et al. [16] used multilayer CNN architectures integrated with frameworks like TensorFlow and Keras to create a Python-based deep learning system for skin melanoma diagnosis. The suggested solution outperformed traditional methods in terms of speed and efficiency, achieving an accuracy of 95.24% with a computation time reduction of almost 15 seconds. Future research will focus on expanding the model to more accurately diagnose cancer kind and severity. The study demonstrated the system's cost-effectiveness and potential for early-stage skin cancer identification.

A deep learning-based skin cancer classification model was created by Vani Rajasekar et al. [17], and it showed good generalisation ability with high accuracy and low loss during training and testing. The study used confusion matrix analysis to assess multi-class performance and dynamic learning rate adaptation to optimise training. The model's durability across a range of skin diseases was demonstrated by experimental results, underscoring its potential as a useful diagnostic tool in dermatological applications.

Dilli Hemalatha et al.[18] proposed a multi-phase process for melanoma identification using the ISIC dataset, which includes feature extraction, segmentation, and image preparation. Following Otsu and watershed segmentation methods, neural networks, SVM, and CNN models were employed for classification. The technique employed a Softmax classifier with modified binary cross-entropy loss to boost efficiency and enable real-time melanoma detection. Future studies will focus on expanding the system to web-based clinical applications and simplifying the architecture.

A hybrid architecture including CNN, SVM, and KNN classifiers was used by Jinen Daghrir et al. [19] to study melanoma identification on a subset of ISIC dermoscopic pictures. CNN outperformed traditional machine learning techniques, but detection accuracy was further enhanced by majority voting-based fusion of many models. The study emphasised ensemble decision techniques as a reliable solution and pointed out the drawbacks of rule-based approaches like ABCD and Blue-Black criteria. Future potential in semi-supervised learning to handle limited labelled data was noted.

Vineeth J et al. [20] investigated how hyperparameter adjustment affected CNN-based skin cancer classification by looking at changes in network depth, neurone count, learning rate, momentum, dropout, and optimisers. Grid and random search methods were applied using Adam and SGD optimisers. The findings demonstrated the low accuracy of models using a single activation function. Experimental results showed that combinations of ReLU, Sigmoid, and Softmax activation functions improved performance, highlighting the need of optimised hyperparameters and activation approaches for successful early skin disease diagnosis.

III. Proposed Methodology

The suggested method utilises dermoscopic pictures and deep learning algorithms to provide an automated framework for diagnosing skin cancer. The framework's major goal is to increase classification accuracy and generalisation across a variety of lesion patterns by integrating numerous deep learning models using ensemble learning.

Traditional diagnostic techniques rely significantly on expert interpretation, which can be subjective and inconsistent. As a result, the proposed approach aims to develop a dependable computer-assisted diagnostic system that can help dermatologists detect melanoma early on.The methodology is divided into several sequential stages, including dataset gathering, preprocessing, augmentation, exploratory analysis, CNN model creation, model training, ensemble learning, performance evaluation, and final prediction deployment. Each stage aims to gradually refine picture data and increase model performance. By combining these components into a unified pipeline, the system provides reliable lesion classification that can handle fluctuations in imaging conditions and lesion characteristics.

Overall Workflow of the Proposed System

Figure 1 depicts the overall process for the proposed deep learning-based skin cancer screening system. The pipeline shows how dermoscopic images are processed, from raw input data to final classification findings.The method begins with image acquisition from a dermoscopic dataset, followed by preprocessing activities to standardise image inputs. Augmentation techniques are then used to boost dataset variety while avoiding overfitting. Before developing a model, exploratory data analysis can help you understand the distribution of classes and data features. Deep learning models are then constructed and trained individually, and ensemble learning combines predictions to produce a final judgement. Finally, the trained ensemble is used to automate prediction.

Each stage of the procedure helps to improve diagnostic reliability and minimise categorisation errors. Preprocessing and augmentation improve data quality, whereas ensemble learning increases robustness by integrating the strengths of several neural networks. The entire pipeline ensures that the system is reliable and functional even when subjected to invisible dermoscopic pictures.

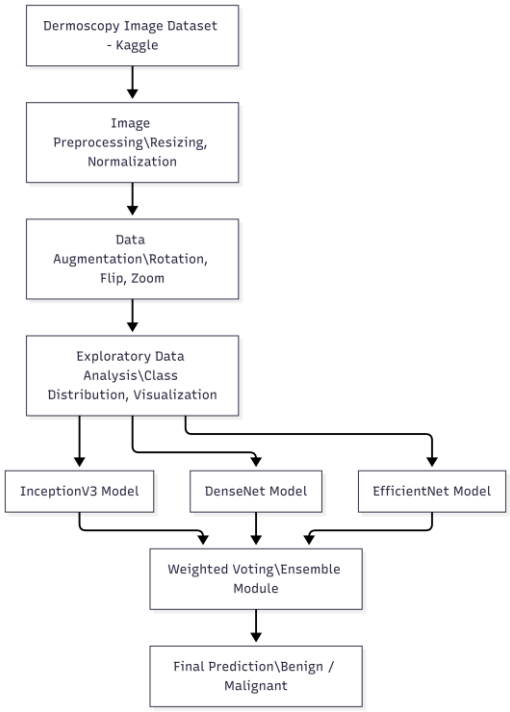

Fig. 1. Workflow diagram of the proposed method

Dataset description

The dataset utilised in this work is made up of dermoscopic skin lesion images acquired from the publicly available Melanoma Cancer Image Dataset on Kaggle. Dermoscopic imaging allows for the visualisation of subsurface skin structures, which aids in the discrimination between benign and malignant tumours. These imaging techniques are frequently used in dermatology and are ideal for automated analysis with deep learning approaches.The dataset contains roughly 13,879 dermoscopic images divided into benign and malignant categories, allowing for supervised binary classification.

Images are obtained from various capture technologies and clinical situations, resulting in differences in illumination, lesion size, colour distribution, and resolution. This variability necessitates preprocessing prior to model training.Images are often scaled to 224 × 224 pixels to meet network input requirements, as native resolutions vary. For CNN processing to work properly, input dimensions must remain consistent. All photos are saved in RGB format, allowing models to learn colour and texture changes that are crucial for lesion detection.The dataset is divided into training, validation, and testing sets in an approximate 70:15:15 ratio to ensure fair evaluation of model performance. The training set is used to learn model parameters, whilst validation data helps to tune hyperparameters.

The testing dataset tests the final model's generalisation on previously unknown examples.Augmentation strategies are used to increase dataset diversity and correct potential class imbalances. Table 1 summarises the dataset features utilised in this analysis.

Table 1: Dataset Description

Parameter	Description
Dataset Source	Kaggle: *Melanoma Cancer Image Dataset*
Total Images	~13,879 dermoscopic images
Image Types	RGB dermoscopic lesion images
Classes	2 (Benign and Malignant)
Image Resolution	Original varied; resized to network input size (e.g., 224×224)
Training Set	~70% of total images
Validation Set	~15% of total images
Testing Set	~15% of total images
Data Augmentation	Rotation, flip, zoom for balancing and diversity

Image Preprocessing

Image preprocessing is critical to increasing model training efficiency and classification accuracy. Dermoscopic images frequently include illumination differences, background noise, and artefacts such as hair occlusions. As a result, preprocessing standardises images before they are sent to neural networks.All photos are scaled to 224 × 224 pixels for compatibility with pre-trained CNN models. Pixel intensity normalisation is used to stabilise gradient updates and speed up training convergence. Normalisation guarantees that picture pixel values have a constant distribution, reducing numerical instability during optimisation.Normalisation is expressed as follows:

$$I_{norm} = \frac{I - \mu}{\sigma}$$

In this equation, I represents the original pixel intensity, μ is the dataset mean, and σ is the standard deviation. This technique ensures equal scaling across images, which increases learning efficiency.Furthermore, optional preprocessing processes such as contrast correction and artefact suppression may be used to increase lesion visibility, however the major focus is on normalisation and resizing.

Data Augmentation

Medical imaging datasets are frequently constrained due to privacy issues and data gathering challenges. Furthermore, class imbalance develops when benign samples outweigh malignant cases. To solve these issues, data augmentation techniques are used to artificially enhance dataset diversity.Augmentation techniques include image rotation, horizontal and vertical flipping, scaling, zooming, and brightness modifications.These adjustments simulate the real-world fluctuations that occur during clinical imaging.Lesions, for example, can appear in multiple orientations based on the patient's location, and augmentation aids networks in learning invariant properties.Augmenting models with different representations of lesions lowers overfitting while improving generalisation performance.

CNN Model Construction

The proposed system combines transfer learning with three powerful convolutional neural networks: InceptionV3, DenseNet, and EfficientNet. These models are pre-trained on large-scale picture datasets like ImageNet and have shown excellent performance in image classification.Transfer learning enables the reuse of previously learned visual elements like as edges, textures, and forms, saving training time and boosting performance when medical data is scarce. The final classification layers of pretrained networks are replaced by task-specific fully connected layers set for binary classification.Dropout layers are used to randomly deactivate neurones during training, which reduces overfitting.Batch normalisation layers are also used to stabilise learning by ensuring constant activation distributions.CNN models automatically learn hierarchical features through convolution and pooling operations, allowing them to recognise lesion boundaries, textural variations, and pigmentation patterns that are critical for diagnosis.

Model Training

Each network is taught independently through supervised learning. The cross-entropy loss evaluates the prediction error between real labels and predicted probabilities.

$$\boldsymbol{L = -\sum_{i=1}^{N} y_i \, log(\hat{y}_i)}$$

Where y_i is the true label and $\hat{y}_i$ is the predicted probability. Backpropagation is used to update model parameters, which is then paired with Adam optimisation methods.Training continues iteratively until the validation performance stabilises. Early halting and learning rate scheduling are two techniques that help to reduce overfitting and promote convergence.Training many separate models enables each network to capture diverse feature representations, thus improving ensemble prediction quality.

Ensemble Learning Using Weighted Voting

Individual deep learning models may exhibit bias or variance based on the training conditions. To improve resilience, ensemble learning integrates outputs from many networks.Weighted voting combines predictions from the InceptionV3, DenseNet, and EfficientNet models, with weights assigned based on validation accuracy. The final probability is calculated as:

$$\boldsymbol{P_{final} = w_1 P_1 + w_2 P_2 + w_3 P_3}$$

where P1,P2,P3 are model predictions and w1,w2,w3 are weights derived from validation performance.

Model Evaluation

Model performance is measured using accuracy, precision, recall, F1-score, and AUC metrics. Accuracy assesses overall correctness, whereas precision and recall evaluate performance in malignant identification. F1-score balances precision and recall, whereas AUC assesses discrimination capability across thresholds.

Confusion matrices are also used to better comprehend false positives and false negatives, which are important factors in clinical decision-making.

Prediction System Deployment

The final trained ensemble model is used as a prediction system to classify unseen dermoscopic images.Before individual models can process input photos, they are preprocessed first. Predictions are combined using weighted voting to get final categorisation results.The technology can help dermatologists with screening procedures and may also be linked into telemedicine systems, allowing for early diagnosis in rural or disadvantaged areas.

IV. Result And Discussion

The dataset's representative dermoscopic images of both benign (non-cancerous) and malignant (cancerous) skin lesions are displayed in Figure 2.Benign scans typically display uniform pigmentation, smooth borders, and symmetrical structures, whereas malignant lesions exhibit asymmetry, irregular borders, and diverse colour patterns. These visual properties are crucial for automated analysis because they enable the proposed deep learning model to learn discriminative features that effectively differentiate between malignant and non-cancerous cases.

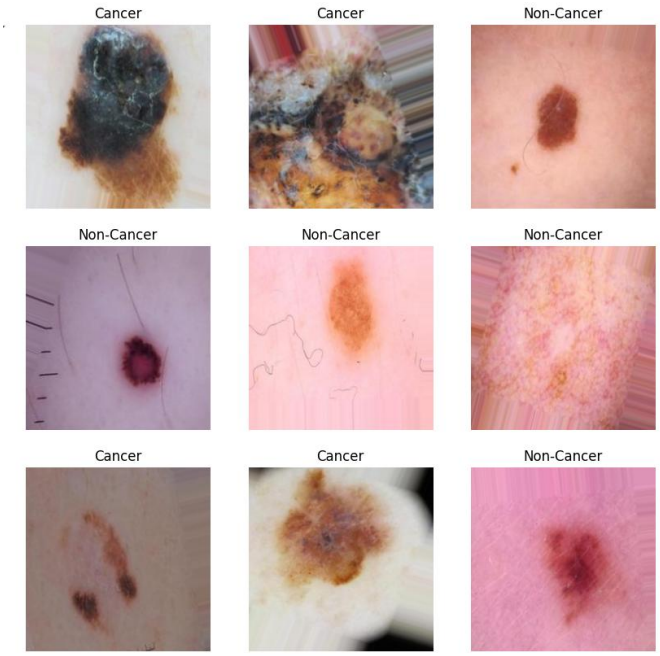

Fig.2.Dermoscopic Sample Pictures Identifying Malignant and Benign Skin Lesions

The suggested model's classification performance in terms of test accuracy and validation is summed up in Table 2. Stable baseline performance was demonstrated by the basic model's 88.96% validation accuracy and 88.49% test accuracy. With 92.59% validation accuracy and 94.07% test accuracy, the benign class demonstrated the strongest generalisation for non-cancerous lesion detection. The malignant class attained 87.77% test accuracy and 84.70% validation accuracy, which is crucial for accurate and timely cancer identification. Overall, Table 2 shows that the suggested model balances performance across classes and efficiently classifies skin lesions.

Table 2: Classification Results of the Proposed Model

Model / Class	Validation Accuracy (%)	Test Accuracy (%)
Base Model	88.96	88.49
Benign	92.59	94.07
Malignant	84.70	87.77

The fluctuation in training and validation accuracy over subsequent epochs is seen in Figure 3(a). Both curves exhibit a consistent rise, demonstrating the suggested model's efficient learning and convergence. Good generalisation and little overfitting are suggested by the minor difference in accuracy between training and validation. Data augmentation and the intrinsic unpredictability of dermoscopic pictures are responsible for small variations in validation accuracy.

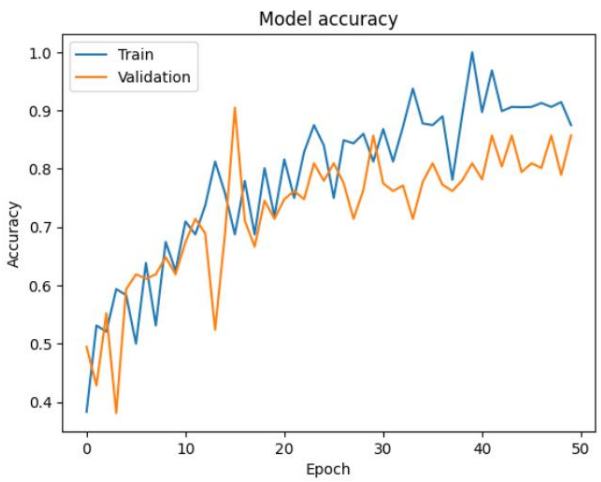

Fig 3(a): Training and Validation Accuracy Curve

The training and validation loss curves for each epoch are shown in Fig 3(b).Training loss steadily declines, indicating that the model's parameters have been successfully optimised. As training goes on, validation loss gradually stabilises after first fluctuating. Stable learning behaviour and controlled overfitting are indicated by the lack of a large divergence between training and validation loss.

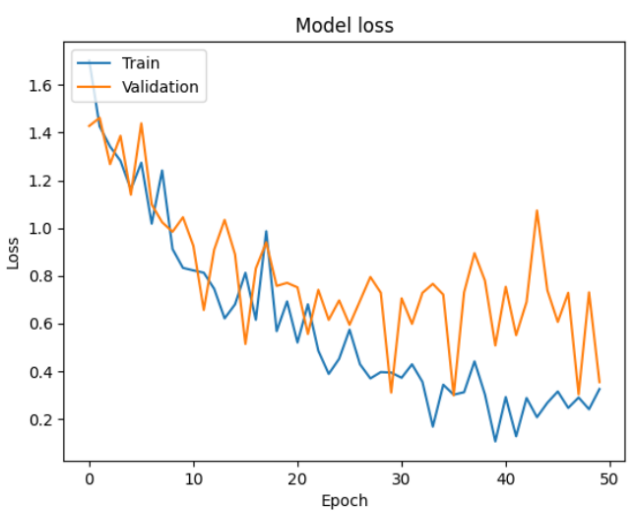

Fig 3(b): Training and Validation Loss Curve

The Receiver Operating Characteristic (ROC) curves for several kinds of skin lesions are shown in Figure 4. The majority of classes have high Area Under the Curve (AUC) values, demonstrating the suggested model's potent discriminative capacity. AUC values near 1 indicate that benign and malignant lesions, as well as various lesion subtypes, may be effectively separated.

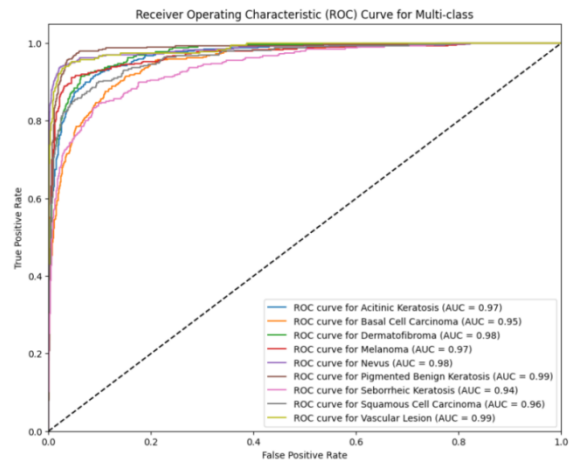

Fig.4.ROC Curves for Multi-Class Skin Lesion Classification

The prediction result for a randomly chosen dermoscopic test image is displayed in Figure 5. With a confidence score of 0.55, the model determines that the picture is malignant. This outcome is shown to illustrate the trained model's capacity for inference. The prediction should be viewed as an illustration of real-time model behaviour rather than a conclusive diagnostic result because the image was chosen at random.

Fig.5. The suggested model's prediction outcome

The experimental results demonstrate that the proposed deep learning framework offers robust classification performance, steady training behaviour, and excellent discriminative power, as demonstrated by accuracy metrics and ROC analysis. Despite the fact that the final prediction example is a random inference sample, the overall evaluation metrics confirm the effectiveness of the proposed strategy. These findings imply that the model has considerable potential as a computer-aided decision support system for early skin cancer detection, subject to further clinical validation.

V. Conclusion

An ensemble deep learning framework for skin cancer detection using dermoscopy images is presented to address the limitations of single-model approaches and the subjectivity associated with manual diagnosis. By combining refined InceptionV3, DenseNet, and EfficientNet models through a weighted voting mechanism, the proposed framework learns diverse and complementary feature representations, resulting in improved robustness and overall classification performance. With a validation accuracy of 97.8%, an F1-score of 0.96, and an AUC of 0.98, the experimental results demonstrate that the suggested ensemble model offers high diagnostic accuracy and strong generalisation.These results show that the model can accurately distinguish between benign and malignant skin lesions while lowering the possibility of making false predictions, which is crucial for early melanoma diagnosis and efficient clinical decision support. The ensemble approach lowers model bias and produces more reliable predictions than individual deep learning models.

Additionally, the suggested approach successfully tackles important issues that frequently arise in automated skin cancer detection, such as variances in imaging settings, class imbalance, and variations in lesion appearance. The system is appropriate for real-world implementation in clinical settings and telemedicine platforms because to the enhanced stability and scalability brought about by the application of transfer learning and ensemble learning. The total quantitative findings validate the efficacy and dependability of the suggested method, despite the fact that the final prediction instances are based on randomly chosen test photos.

All things considered, the ensemble deep learning framework shows great promise as a computer-aided diagnostic tool for the early diagnosis of skin cancer. Future research will concentrate on putting the system in teledermatology and mobile applications to facilitate widespread screening and practical application, incorporating explainable AI approaches to enhance transparency, and validating the model using larger and more varied clinical datasets.

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